Activity 7.1: Area of Polygons & Circles

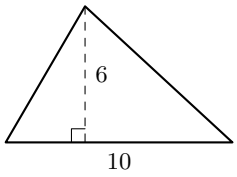
A

Name: _____ Partner's name: _____ Date: _____

Directions: Find the area. Give exact answers (use π and radicals, not decimals). Diagrams are accurate, but use math rather than a ruler or protractor.

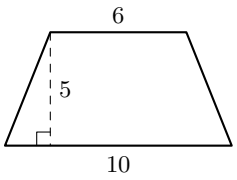
Work alone, then compare after each problem. If answers differ, find the mistake together.

1 Find the area of the triangle.



Answer: _____ Match? ☐ Yes ☐ No

2 Find the area of the trapezoid.



Answer: _____ Match? ☐ Yes ☐ No

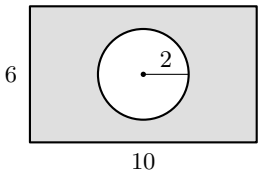
3 Find the area of a parallelogram with base 9 and height 4.

Answer: _____ Match? ☐ Yes ☐ No

4 Find the area of a circle with radius 3.

Answer: _____ Match? ☐ Yes ☐ No

5 Find the area of the shaded region.



Answer: _____ Match? ☐ Yes ☐ No

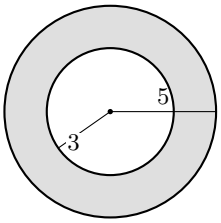
6 Find the area of an equilateral triangle with side length 6.

Answer: _____ Match? ☐ Yes ☐ No

7 Find the area of a rhombus with diagonals 8 and 12.

Answer: _____ Match? ☐ Yes ☐ No

8 Find the area of the shaded ring.



Answer: _____ Match? ☐ Yes ☐ No

Activity 7.1: Area of Polygons & Circles

B

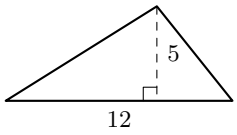
Name: _____ Partner's name: _____ Date: _____

Directions: Find the area. Give exact answers (use π and radicals, not decimals). Diagrams are accurate, but use math rather than a ruler or protractor.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

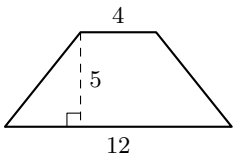
Find the area of the triangle.



Answer: _____ Match? ☐ Yes ☐ No

2

Find the area of the trapezoid.



Answer: _____ Match? ☐ Yes ☐ No

3

Find the area of a parallelogram with base 12 and height 3.

Answer: _____ Match? ☐ Yes ☐ No

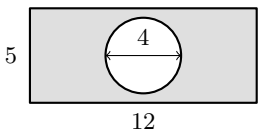
4

Find the area of a circle with circumference 6π.

Answer: _____ Match? ☐ Yes ☐ No

5

Find the area of the shaded region.



Answer: _____ Match? ☐ Yes ☐ No

6

Find the area of an equilateral triangle with altitude $3\sqrt{3}$.

Answer: _____ Match? ☐ Yes ☐ No

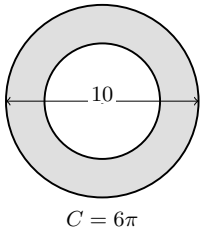
7

Find the area of a kite with diagonals 6 and 16.

Answer: _____ Match? ☐ Yes ☐ No

8

The outer circle has diameter 10 and the inner circle has circumference 6π. Find the area of the shaded ring.



Answer: _____ Match? ☐ Yes ☐ No